MODEL GUIDE

How it works

Choose a route, generate a scenario and compare the fuel plan at each stop.

Scenario data

Airport coordinates come from public geographic data. Prices, supply limits and aircraft parameters are generated assumptions for exploring the model. The same settings produce the same inputs. Replace them with your own data when available.

1 Choose the route

Choose the departure and destination, then add stops to build a continuous route of 1–5 legs. The map follows your choices. The model keeps this route fixed and chooses how much fuel to buy at each stop. An example route is available separately.

2 Generate the inputs

Generate a complete scenario for the route. Airport prices and supply limits come from the scenario rules. Distance and the selected aircraft profile give flight time, fuel burn and reserves. You can then edit the values you want to test.

3 Run the calculation

Run the optimizer after editing. It checks fuel balance, tank capacity, takeoff weight, landing weight, reserves and each airport’s uplift limit. If the route cannot be served with those inputs, the result shows which limits need attention.

4 Compare both plans

Compare buying as needed at every stop with buying extra fuel earlier. Both plans start and finish with the same fuel. Read the purchase cost alongside extra burn, uplift by airport and remaining fuel. Playback shows how those values change along the route.

5 Bring in your own data

Import dated airport quotes, choose the flight date and update the aircraft and supply inputs. Prices use USD per metric tonne with into-plane fees included. Missing or overlapping quotes are shown for correction. Export the inputs and results to keep a copy of the comparison.

How to read a lower cost

A lower purchase cost can come with more fuel burned in flight. That extra fuel is already included in the purchase calculation. Add any separate carbon charges, handling costs or payload effects when assessing the final benefit.

Data to connect next

Where each input comes from and what it changes

Input	Source	Use in the model
Airport fuel prices	Supplier quotes, airline contracts and invoices	Replace generated prices. Keep validity, currency, units, fees and buyer terms together.
Aircraft performance	Airline-approved performance and flight-planning data	Replace burn rates, weight limits, reserves and the extra-weight fuel penalty.
Fuel availability	Suppliers, airport fuel operators and contracted delivery rights	Set usable uplift limits for each visit and test a supply interruption.
Market benchmarks	Public energy prices, exchange rates and national fuel balances	Track the market background and set scenarios. Airport contract prices remain separate inputs.

Try one change at a time

Change one airport price or supply limit, recalculate and compare. Then vary the aircraft or route. This makes it easier to see what is driving the result. The research tab explains the fuel-carriage trade-off in more detail.

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